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B.E. - (Mechanical) (Sem-VII) (Revised Course 2019-2020)

EXAMINATION JANUARY 2024

OPEN ELECTIVE - Additive Manufacturing

[Time: 3 Hours]

[Max. Marks: 100]

- Instructions:**
- 1) Answer the questions from **Part A, B and C** as per the instruction.
 - 2) Assume missing data if any
 - 3) Draw neat sketches wherever necessary

PART A

Answer any **TWO** questions from the following:

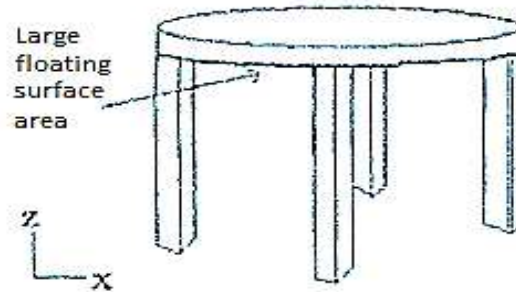
- | | | |
|----|---|-----------|
| Q1 | A. Explain the advantages of AM processes over Conventional processes | 08 |
| | B. Which additive manufacturing process is used by the casting company to make the patterns for investment casting using Liquid thermoset polymer resins? Justify your answer with the working principle, and explain the process | 06 |
| | C. Classify the Additive manufacturing process | 06 |
| Q2 | A. Suggest the additive manufacturing process which can print three Dimensional objects from Titanium material. Explain the working principle, advantages & disadvantages of the Process | 08 |
| | B. Explain the different components of FDM machine | 06 |
| | C. Explain the role of Additive manufacturing in new product development with the aid of flow chart. | 06 |
| Q3 | A. Explain the process, advantages and disadvantages of laminated object manufacturing with neat sketch | 08 |
| | B. Explain the advantages and disadvantages of solid ground curing process | 06 |

- C. Explain the different process parameters which govern the product quality developed by fused deposition technique process. **06**

PART B

Answer any **TWO** questions from the following:

- Q4 A. Explain the importance of proper part build orientation. **06**
 B. Explain the process of reverse engineering with suitable example. What is its role in Additive manufacturing **08**
 C. How do you generate a surface from a point cloud? **06**
- Q5 A. Explain the part building errors in SLS and how to compensate for it? **06**
 B. Differentiate between silicone rubber tooling and aluminum epoxy tooling in detail **08**
 C. Explain the concept of part orientation and support generation for the following 3D part. **06**



- Q6 A. Write a short note on object quadra systems **06**
 B. Explain how Internet based software for RP offers a collaborative production environment among users **06**
 C. Write a note on Customized implants and prostheses made from additive manufacturing process. **08**

PART C

Answer any **ONE** question from the following:

- | | | |
|----|---|-----------|
| Q7 | A. Explain how the various process parameters affects the build time of the product in Additive manufacturing process. | 06 |
| | B. What is the role of additive manufacturing in customized implants and prosthetics | 06 |
| | C. Explain in detail the Data Processing for Additive Manufacturing Technology | 08 |
| Q8 | A. Compare SLS, SLA LENS with respect to following parameters i) laser used ii) material used iii) surface finish iv) support structure | 08 |
| | B. Write a note on Computer Aided Tissue Engineering (CATE) | 06 |
| | C. Explain the different components of FDM machine | 06 |